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**Objective:**

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Obtain a challenging leadership position and utilize my experiences in the **end to end product delivery, product roadmap creation, service & strategy planning, Organizational Development, product/process specifications and product/process driven development** and lean management skills with a growing company to achieve optimum utilization of myself and maximize profits.

Innovative and high performing with a proven record of driving revenue and profit growth for multi-billion-dollar businesses. Skilled at transforming businesses by creating a strong vision, robust strategy, high performing organization and innovating with technology.

- I am a passionate, proactive, results-driven professional, having industrial experience in project management, product development and digital transformation. I have delivered various innovative product in global market.
- I have established strategic R&D program and have deeper understanding of business models, revenue analysis and current technology trends, having experience and understanding in the advanced technology areas such as Artificial Intelligence, Cyber Security, and Internet of Things (IoT). Exposure to Cloud, SaaS, and emerging web/mobile and enterprise platforms.
- I have experience in building consumer-centric industry solutions coupled with the go-to-market, business development, and marketing strategy. Entrepreneurial mindset, resolving business challenges, turning ideas into reality, proven track record in taking innovation to commercialization, facilitating growth and driving real time impact.

• Project Management • Project Planning • Finance • Operational Metrics • Resource Management • Process Improvement • Budgeting & Forecasting • Stakeholder Engagement • Product Development (Agile, Scrum, Kanban) • Innovation • Business Development • Consulting • Market Research

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**Professional Profile and achievement:**

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- Overall **21+** years of experience in **technically managing and End-End Product delivery. 10+ years' On-site (6+ years in USA, 3 year in Germany many European countries and 1 + years in Japan)** and **11+ year's offshore India** management and end to end Products, Tools and hardware component development and delivery experience.
- Manage a team of on-roll employees and off-roll employees, having good team handling and mentoring experience of 5 to 76 employees.
- Interfaces with multiple international clients, stakeholders for smooth execution of overall program. **8+ years Semiconductor** domain and **6+ years of automotive ADAS** domain and **7+ Years in Locomotive/Railway domain** experience for end to end execution of product portfolio and its delivery to multiple clients around the world.
- Demonstrated success defining and launching large scale product & service platforms.
- Significant experience in the **product roadmap, service planning & strategy, Organizational Development, Resources Management & Training.**
- Good exposure on Design product requirements based on **business** and **technical** needs, write **product/process specifications** and **drive product/process development.**
- Knowledge of commonly used collaboration tools supporting agile process like JIRA, VersionOne, story boarding.
- Experience in **End to End Product delivery, Requirement gathering, analysis, Design, Coding, Hardware-Software Integration, Software-Software Integration and Project and Product Management.**
- Extensively used **Agile for 10+ years** and Waterfall Model for my rest exp, Worked as Scrum Master for couple of projects as well as Product owner for one product.
- **Planning, Risk Management, finalizing the scope of work** and be a part of designing/developing Applications, prototypes, or proofs of concepts (POC's) Gathering functional requirements, developing technical specifications, and project & test planning.
- Experience in implementing process Software Process Improvement and Capability Determination (SPICE) and practices in software development life cycle.
- Good understanding of software framework, products, product technology and work practice.
- Ability to quickly understand, learn and work in various technical environments, Strong Experience in programming fundamentals.

- Experience in working as Product Owner and the team to create backlog. Prioritize and assign complexity points to the user stories based on its business/client deliverable value.
- Report team progress to stockholders by burn-down charts, dashboards reporting and sprint release progress using -VersionOne tool and very well versed with **Project Management Framework, Project Management Processes**.
- Possess critical abilities such as thinking strategically, as well as tactically, and ability to successfully prioritize tasks in order to ensure that all projects are on schedule
- Proven initiative and flexibility, detail oriented, and ability to effectively work on multiple projects simultaneously
- Excellent teamwork skills including the ability to establish cross-functional, collaborative relationships and conflict management.
- Excellent written and verbal communication and interpersonal skills with ability to work effectively with multiple projects.

Being **creative and research oriented imaginative thought** during my course in **Indian Institute of Science** I developed two products for **IISc** with a team of five people; one is for MEMS group and other one for Electronics and Communication group.

### Education and Personal Achievement

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**Google Hackathon:** I got Second Prize from **Google and Indian Institute of Science Hackathon**.

**Swiss Bank:** I got prestigious "**Swiss bank gold key**" member award for best Technical product in IT solution.

**Patents:** I have **four patents** in the areas of Encoding and compression Algorithm, Automotive and Metro-Locomotive.

**Publication:** I have published and presented multiple papers with in country and outside country.

**Mobile and Web Application:** I have **five android based mobile applications and five live web sites**.

**Start-Up:** Co-Founder and VP of DOILIVE, First Indian startup company to work with online Doctor and Patient video call facilitator with 35 employee and after three+ years, sold company to Dubai based YCWC company with 36 Cr.

**Social Work:** I have contributed my technical expertise suggestions for various critical projects in IISc.

**Award and Recognition in IT:** I have earned **9 awards** from organizations I have worked on. I have written one book for **Computer handling and diagnostic for students and engineers** and that is available in market and sold by Media Fusion Private Ltd.

**Certification:** Earn Two Certifications from **Indian Institute of Science and PEP and Scrum Certified**.

### Technical and Managerial skills

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- Ability to direct and manage complex product development projects to successful on-time completion.
- In-depth knowledge of fundamental engineering principles.
- Strong engineering process development background.
- Experienced with product development to meet industry standards such as ASPICE or CMMI.
- Effective interpersonal skills with customers, peers, leaders, global teams, and industry forums.
- Desire to maintain skills and master new technologies.

**RESPONSIBILITY PERFORMED:** Have earns various Technical and Management skills as Program Manager, Technical Architect, Scrum Master, Product owner and Technical Leader represent a unique combination of me. Personally, I have derived and determination to consistently achieve success as per my role for the organizations that I have worked with in the past.

### COMPANY WORKED ON:

1. **ALSTOM Transport India Ltd. [Location: India ] Since 2017**
2. **Continental Automotive India Pvt Ltd [Location: India ] [Duration: 2014 to 2017]**
3. **HCL America Inc and HCL India [Location : USA, Japan and India] [Duration: 2009 to 2014]**
4. **Indian Institute of Science sponsored by Cranes Software International Ltd. [Location: India ] [Duration: 2003 to 2009]**

### Project Experience Profile

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**Current:- Automatic Train Supervision (ATS)****Dec 2017 to Till Now****Team Size:** 15**Company:** ALSTOM Transport India**End to End Management:** End to end Delivery of Urban Solution for **Turin L-1 [Italy], SGP L-18 [France], Lyon [France], Tel [Singapore], SSL6 [Singapore], and GTCS Canada Hybrid Train project.****Responsibility:**

- ☐ Collaborated with multiple international clients for end to end solution delivery—Some of it are **Turin L-1 [Italy], SGP L-18 [France], Lyon [France], Tel [Singapore], SSL6 [Singapore], and GTCS Canada Hybrid Train project** —across the EU, Singapore and CANADA regions.
- ☐ Worked closely with key Indian stakeholders from **Mumbai, Delhi, Bangalore, Kochi, and Lucknow** to define roadmaps and ensure delivery aligned with contractual commitments for metro agreed solution as per tender.
- ☐ Oversee key project management various streams including, **System to sub-system and Interface Management, Scope Management, Schedule Management, and Resource Management.**
- ☐ Managed end-to-end delivery of the **Automatic Train Supervision (ATS)** with Fluence System to various clients.
- ☐ Communicated effectively with diverse **international and external stakeholders** to ensure alignment and smooth execution.
- ☐ Led comprehensive project planning and execution activities.
- ☐ Ensured smooth execution and delivery of mutually agreed Fluence solution [Train to Train Communication] to customers through effective coordination and stakeholder management.

**Previous A-1:- An IoT application software for Hardware In Loop (HiL)(ADAS)****Oct 2016 to Nov 2017****Team Size:** 10**Company:** Continental Automotive Inc., Germany**Responsibility:** R&D on new features and POC creation along with Managed Engineering product.

- ☐ Managed the end-to-end development and delivery of the **HiL (Hardware-in-the-Loop) Management Tool.**
- ☐ Designed a **three-tier architecture** for the product, led its technical design, and mentored team members to ensure on-time delivery.
- ☐ Reported team progress to stakeholders using **burndown charts, dashboard reporting, and sprint/release tracking** through VersionOne.
- ☐ Implemented **Raspberry Pi-based hardware control**, enabling diagnostics and command execution across multiple hardware modules.
- ☐ Guided the strategic transition of the product to a **SaaS-based cloud architecture**; after evaluating Amazon, Google, and Microsoft cloud platforms, selected **AWS** based on technical requirements.
- ☐ Designed and implemented a **scheduler service** running on the IoT server, along with distributed services on each HiL system for seamless communication between devices and the IoT platform.
- ☐ Mentored junior engineers and new team members, providing technical leadership and solving complex programming and design challenges.

**Previous A-2: - Automation and IoT application for HiL and various sensors MFC, MFL and ARS (ADAS)****Aug 2016 to Nov 2017****Team Size:** 21**Company:** Continental Automotive Inc., Germany**Technologies Used:** CAN protocol, Vector CANoe 8.2, CAPL Scripting VC++, MFC, and C++, Java Scripting, Python and batch scripting.

**Responsibility:** Managed end to end product called HiL Automation Tool. Created two tier architecture for HiL Automation Tool, Designed this application that can communicate and control the software and various hardware communications such as ECU, Image Emulator and Bus Emulator with help of relay board. Completed the development and mentored team members to deliver the product on time. This includes my responsibility for reporting team progress to stockholders by burn-down charts, dashboards reporting and sprint release progress.

My responsibility for this product also includes the designing of new measurement objects that extended the product capability by 300% and increased the usage percentage of this product by three folds. After increasing the capability of this tool, we have received multiple appreciation by various customers and counterparts for this product and gain of more customers that intern increases the sale of Monitor HiL double within 2 quarters. This increases the revenue of project by 60-70%.

**Previous A-3:- Raw Data HiL software for various sensors MFC, MFL and ARS (ADAS)****July 2016 to Nov 2017****Team Size:** 21**Company:** Continental Automotive Inc., Germany**Technologies Used:** CAN protocol, Ethernet protocol, Vector CANoe 8.2, VC++, WTL, Boost, and C++, Java Scripting, Python and batch scripting.

**Responsibility:** Managed end to end product called RDH Tool. Created desktop application that takes input as a recording file and apply multiple algorithms to segregate the image data and bus data that will be send to Image Emulator and Bus Emulator respectively via Ethernet for SMFC4B0 ECU RDH and via CAN protocol for MFC431 ECU RDH. This application is design in such a way any diagnostic information can be caught for analyzing the issue on framework. Completed the development and mentored team members to deliver the product on time.

This includes my responsibility for reporting team progress to stockholders by burn-down charts, dashboards reporting and sprint release progress. This tool is one of the finest product and best HiL solution as of now. We have received multiple appreciations by various customers and counterparts for this product and gain of more customers that intern increases the sale of Raw Data HiL double folded within 3 quarters. This increases the revenue of project by 40-50%.

#### Previous A-4:- ADTF Application development for BMW (Advance Driver Assistance System (ADAS))

March 2015 to Dec 2016

**Team Size:** 21  
**Company:** Continental Automotive Inc., Germany  
**Technologies Used:** C#, VC++, WTL, and C++, Boost, CAN, Flexray and Ethernet Protocol.

Developing new features and supporting issues for ADTF (Automotive Data and Time Triggered Framework) that is widely used by various automobile companies e.g, Audy, BMW and Daimler etc. Main work of the application is to get the data from various sensors eg. MFC (Multy Function Camera), ARS (Automotive Radar Sensors), GPS (Global Positioning System) through various protocol such as CAN, Ethernet and Flex Ray protocol and plot the data graph as per customer need, e.g, 2D Graph, Tabular data etc.

**Responsibility:** Involved in Architecture modification and design update and development of the product.

#### Previous A-5:- MTS (Measurement Testing and Simulation. (ADAS)

Des 2014 to Mar 2016

**Team Size:** 21  
**Company:** Continental Automotive Inc., Germany  
**Technologies Used:** VC++, WTL, and C++, Python.

Developing new features and supporting issues for framework called MTS (Measurement Testing and Simulation Framework) that is widely used by various automobile companies e.g, Audy, BMW and Daimler etc. Main work of the application is to get the data from various sensors eg. MFC (Multy Function Camera), ARS (Automotive Radar Sensors), GPS (Global Positioning System) through CAN, Ethernet or Flex Ray protocol and plot the data graph as per customer need, e.g, 2D Graph, Tabular data etc.

**Responsibility:** Involved in Architecture modification, Design update and Mentoring Team to accomplish the on time delivery of the product..

### 3. OBSx suite of Banking Application (Interest Rate and Derivative)

Feb 2012 to Aug 2014

**Team Size:** 27  
**Client:** UBS Inc., USA (Company Name: HCL America Inc)  
**Technologies Used:** Objective C, C# and C++, Sybase, Python, Tcl and Objective C++, Mac OS, Share Point, Linux and NeXT OS, .Net.

As part of OBSx suite of application, which I provide development and supports are contain highly confidential data that are developed on MAC OS and use production database that are available in Chicago, New York and Stamford USA

1. PantherNT : This application is use for end of the day trading risk and PnL reporting in the bank and use of this application is restricted within bank as it contains confidential data. This application made up with the help on three components. GUI is develop in Java and supported for Excel Add-In.

2. OBSX Server: This application is middle layer application and being used by Depo Tool to save any depo trade booked by trades. Hence contain all the trading information for deposit trades.

3. Pricing Server: This application is use for pricing any trade booked by Boomerang application and pricing curve being booked by Catapult.

4. MMF Poller: This application is use for middle layer data massaging so that data can be further use by other UBS Internal applications.

5. Report Batch: This application is heavily used by bank for generating four kind of trading statistics.

6. Edit Screen: This application is developed on NeXT operating system that is very rear now. This application is use for booking any kind of trade like swap, swaption, CMT, FRA, LDC bond and Instrument.

7. STA Reporting: This application is use for generating hourly basis report and contains highly sensitive data for CTC group in the bank.

**Responsibility:** As a team lead and Architect, involved in Architecture modification and design update and development of the product and reporting the project status to Project Manager and other stack holders.

#### 4. MOCVD Application (Metal Organic Chemical Vapor Deposition)

Dec 2010 to Feb 2012

**Team Size:** 30  
**Client:** VEECO Inc., USA (Company Name: HCL America Inc)  
**Technologies Used:** C#, .Net, WPF, WCF, CCF Framework, SQL Server 2008, Telerik control.

MOCVD (Metal Organic Chemical Vapor Deposition) Galaxy software is being used to manufacture LED using cluster tool. I have involved in this product development from beginning. We started this development with CCF framework. We have written the entire module for this software. There are three main modules

In MOCVD production of GaN (Gallium-nitride) -based materials for FETs or LEDs, the K465 GaN offers a unique set of features and functions. The K465 GAN is divided into separate modules, assembled separately on separate wheeled frames, then mounted together, placed on metal feet and the casters are removed. The modules includes Electrical/Circuit Breaker Assembly (CBA) , Electronics EE Rack (for common subsystems)/Power Supply Module, PMA/Source Delivery Assembly (SDA) and Flow Control Module , PMA/Reactor Module (RHA) etc.

The SDA and RHA are housed in a single PMA frame, accessed through separate sets of doors. The Source Delivery Assembly/SDA (for hydrides, alkyl bubbler networks, etc.) collects gas, controls gas, and injects gas into the reactor. The Source Delivery system includes:

- Gas purifiers
- Valves (pneumatic valves, manual valves, needle valves)
- Mass Flow Controllers –regulates (control) the amount of gas flow in the system
- Pressure Controllers - regulates pressure in various portions of the system
- Pressure Transducers and Gauges –used to measure pressure
- Binary Gas Monitor - measures concentration of source materials in gas stream

The components are configured in the form of slices to run the various sequences for flow of gases set by the user. Material deposition occurs through applied gases and metals, heat, carrier rotation and sub-atmospheric pressure. Gases are delivered to wafer surfaces in the Reactor Housing Assembly/Growth Chamber. The heater component in the reactor is categorized into various filaments to cause uniform heating at various temperatures. The further process involves cooling and dicing of wafers. This system constitutes the core of MOCVD process.

1: **Operator Interface-** OI we had developed in **C#** forms but gradually we opted to have good screens hence opted for **WPF** technology and converted the screens to WPF screens

2: **Supervisor-** Heart of application and all of the business logic is being developed to communicate with OI and HW components.

3: **Simulator-** Since offshore team does not have any hardware hence we had developed this simulator to simulate our software.

**Data Base-** SQL server data base is being used for storing data during the deposition of several layers in process of LED manufacturing.

This is being developed using C#.

#### **Responsibility:**

- Architect, High level and low level design of galaxy software.
- Involved in critical module implementation and end-to-end testing using NUnit.
- As scrum master involved with task allocation, team status tracking, handling standup meeting.

Code reviews with code-collaborator and integration testing with NUnit.

#### 5. Virtual Part Identifier (VPI)

June 2010 to Nov 2010

**Team Size:** 6  
**Client:** VEECO Inc., USA (Company Name: HCL America Inc)  
**Technologies Used:** ASP.NET, WPF,WCF, C#, VB.NET, SQL Server 2008, Solidwork and E-drawing.

This is a web based tool and is being developed for MOCVD field service engineer uses. This tool will help user to identify the MOCVD parts and provide functionality to assemble and disassemble the tool as well as sub systems of it.

This is being developed using client-server model. This VPI server will hold the VPI data and VPI client can browse this for extracting the part information. *Being Product Owner, I was involved on requirement gathering and responsible for module developing and delivering direct to client.*

**Responsibility:** Involved in Architecture modification and design update and development of the product.

#### 6. Sony Bravia DTV Application

March 2009 to June 2010

**Team Size:** 14  
**Client:** SONY, JAPAN (Company Name: HCL Technology Ltd)  
**Technologies Used:** CentOS, C++, C, xml, QT, STL.

Being Team Lead, I was involved and responsible for module name Gemstar EPG developing and delivering direct to client. It's advanced concept of EPG that have rich GUI for better user interaction along with other many new features like, Creating your own flavour profile, that will help you to avoid massive search. Apart from this there are various other features along with very rich database for fast and better scheduling. More can be seen live on Sony Bravia TV features.

Below are the few features.

- **TV Listing:** List of show that will have it for the day.
- My TV: Will have detail of current as well as next program with small clip.
- **Browser search:** It will give you an option to do the various searches according to specific choice like, "Sports", "Movies", "Children", "Educational", "News".
- **My Schedule:** Will have Series manager, reminders, recording etc

#### Responsibility:

- Architect, High level, low level design of Gemstar EPG software.
- Involved in critical module implementation and end-to-end testing.
- Task allocation, team status tracking, handling status meeting.

Code reviews and integration testing.

#### 7. Outlook Sales Force Connector

Jan 2008 to Feb 2009

**Team Size:** 10  
**Client:** Sales Force, USA (Company Name: Cranes Software International Ltd)  
**Technologies Used:** Window platform, C++, C, xml, VC++, MFC, ATL, STL, COM, Microsoft Outlook, SQL Light and VSS.

Outlook Sales Force Connector is a MFC ATL application that will be responsible for communicating between Outlook and Sales force calendar and synchronizing the calendar events from both sides. With the use of this software one can synchronize outlook calendar normal events, recurring events and group recurring events without any discrepancy.

**Responsibility:** As a Team Lead involved in Architecture and design update and development of the product along with reporting the progress of the project to senior management and client.

#### 8. Xtelligent DataStreamer, DataLoad product development.

Sep 2006 to Jan 2008

**Team Size:** 13  
**Client:** Dendrite, USA (Company Name: Cranes Software International Ltd)  
**Technologies Used:** Window platform, C++, C, STL, xml, VC++, MFC, ATL, COM, Microsoft Outlook, SQL Server and Surround SCM.

Involve in Designing and developed of many functionality and architecture of the product as Sr. Software engineer. More description can be seen extended part of resume.

**Responsibility:** As a Team Lead involved in Architecture and design update and development of the product along with reporting the progress of the project to senior management and client.

#### 9. TARANG Audio Player and TARANG Streaming.

Oct 2005 to Aug 2006

**Team Size:** 7  
**Client:** Equabe Communication, India (Company Name: Cranes Software International Ltd.)  
**Technologies Used:** Window platform, C++, STL, C, xml, VC++, MFC, ATL, COM, Microsoft Outlook, SQL Server and Surround SCM.

Involved in full product development from scratch along with HLD and LLD. TARANG - The Audio Codec. TARANG is a new audio codec solution. The algorithm is unpublished and to be patented. The main features of TARANG are:  
 Very low complexity encoding and decoding (small program memory footprint), Encoding speed - five times faster than real-time, Decoding speed - three times faster than real-time, Amenability to mixed signal processing in hardware,

**Responsibility:** Involved in Architecture modification and design update and development of the product.

#### 10. Syatat-12, Sigma Plot 9.0, Table Curve 2D and Table Curve 3D Product.

Feb 2004 to Sep 2005

**Team Size:** 120  
**Client:** Cranes Software International Ltd.  
**Technologies Used:** Window platform, C#, C++, STL, C, xml, VC++, MFC, ATL, COM, SQL Server and VSS.

Being software engineer as a part of big team, I have added one new feature for Systat Software, Due to that feature client can easily use the Systat command prompt .without knowing any command because user can get command prompt help, Just like Microsoft Visual Studio.net. Either you know command or not just press any key then you will get drop down list which will contain all the command related to that class .this I developed by using MFC and VC++. Likewise you will get the help for Argument list, Subcommands, Second level commands and Top Level Commands and second level commands can get help from its own Top level Commands

**Responsibility:** Involved in development of the product.

**Reference**

**Dr. KVS Hari** (IISc Scientist and Chairman of IEEE)

**Dr. Chandrashekhar** (IISc Scientist)

**Dr. Rudra Pratap** (IISc Scientist and Chairman of MEMS and Nano Science)

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